

Graduate Student, MIT Chemistry
Kiessling and Shalek Labs

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EDUCATION

Indian Institute of Science Education and Research, Kolkata

Kolkata, India

B.S.-M.S. Chemistry, 2016-2021

Thesis: "CsPbBr₃ Perovskite Nanowires: Synthesis, Optical properties and Single Particle Tracking"

Advisor: Prof. Prasun K. Mandal (Chemistry)

RESEARCH AND TEACHING EXPERIENCE

Graduate Research Associate

2021-Present

Prof. Alex K. Shalek and Prof. Laura L. Kiessling

Massachusetts Institute of Technology, MA

Developed novel platforms for multimodal single-cell and spatial profiling of functional glycosylation, protein expression, and mRNA levels in human tissues, leveraging DNA-barcoded human recombinant lectins. Utilized these platforms to uncover changes in functional lectin-binding glycan patterns associated with immune activation in peripheral blood, as well as cancer-associated, distinct glycan niches in follicular lymphoma with diagnostic value.

Graduate Teaching Associate

2021-2022

Thermodynamics I (5.601): Prof. Robert G. Griffin and Prof. Brett McGuire, MIT

Thermodynamics II (5.602): Prof. Keith Nelson and Prof. Alex K. Shalek, MIT

Graduate Research Associate

2019-2021

Prof. Prasun K. Mandal

Indian Institute of Science Education and Research, Kolkata

Synthesis, optical characterization and single particle tracking with quantum dots and nanowires

Undergraduate Summer Research Associate

2019

Prof. Lucio Frydman

Weizmann Institute of Science, Israel

Design of pulse-sequences for diffusion-weighted MRI and solid-state NMR

AWARDS AND HONORS

Koch Institute Image Award, MIT, 2026

Neil and Anna Rasmussen Fellowship, MIT, 2025

Koch Institute Image Award, MIT, 2025

David H. Koch Fellowship, MIT, 2024

Howard O McMahon Fund for Physical Chemistry Fellowship, MIT, 2024 (summer)

Owen Fellowship, Johns Hopkins University, 2021 (Declined)

Cornell Fellowship, Cornell University, 2021 (Declined)

Kupcinet-Getz International Summer School, Weizmann Institute of Science, 2019

INSPIRE Fellowship, Government of India, 2016-2021

GRANTS

Coca-Cola Alternatives to Sugar Grand Challenge (PIs: Alex K. Shalek, Laura L. Kiessling), \$650,000, Co-author and Key Personnel, MIT, 2024-2026

Koch Institute Traditional Bridge Award (PIs: Alex K. Shalek, Laura L. Kiessling, Sizun Jiang), \$800,000, Co-author and Key Personnel, MIT, 2024-2026

MIT School of Science Research Innovation Award (PI: Alex K. Shalek), \$150,000, Co-author and Key Personnel, MIT, 2024-2026
 Koch Institute Frontier Award (PIs: Alex K. Shalek, Laura L. Kiessling), \$150,000, Co-author and Key Personnel, MIT, 2023-2024

PUBLICATIONS

* Denotes equal authorship

3. Yiu, S.P.T.*, Chang, Y.*, Yeo, Y.Y.*, Qiu, H.*, Wu, W.*, Michel, H.A., Jin, X., Huang, R., Kure, S., Parmelee, L., Luo, S., Cramer, P., Lee, J.L., Wang, Y., Zhao, Z., Yeung, J., El Ahmar, N., Simsek, B., Mohanna, R., Van Orden, M., Lu, W.S., Livak, K.J., Li, S., Gao, C., Burgess, M., Keane, C., Shahryari, J., Kingsley, L.G., Al-Humadi, R.N., Nasr, S., Nkosi, D., Sadigh, S., Rock, P., Frauenfeld, L., Kaufmann, L., Zhu, B., Basak, A., Dhanikonda, N., Chan, C.N., Krull, J., Cho, Y.W., Chen, C.Y., Brown, J., Wang, H., Zhao, B., Lee, J.J., Loo, L.H., Kim, D.M., Boussiotis, V.A., Zhang, B., Wei, K., Shalek, A.K., Howitt, B.E., Signoretti, S., Schürch, C.M., Hodi, F.S., Burack, W.R., Rodig, S.J., Ma, Q., and Jiang, S. “Same-Slide Spatial Multi-Omics Integration with IN-DEPTH Reveals Tumor Virus-Linked Spatial Reorganization of the Tumor Microenvironment”, *Cancer Discov*, **24**:10.1158/2159-8290.CD-25-0775 (2026)
2. Lensch, V., Gabba, A., Hincapie, R., Bhagchandani, S., Basak, A., Alam, M.M., Noble, J., Irvine, D.J., Shalek, A.K., Johnson, J.A., Finn, M.G., and Kiessling, L., “Carbohydrate-Lectin Interactions Reprogram Dendritic Cells to Promote Type 1 Anti-Tumor Immunity”, *ACS Nano*, **18**, 39, 26770–26783 (2024).
1. Otikovs, M., Basak, A., and Frydman, L., “Spatiotemporal encoding MRI using subspace-constrained sampling and locally-low-rank regularization: Applications to diffusion weighted and diffusion kurtosis imaging of human brain and prostate”, *Magn. Reson. Imaging*, **94**, 151-160 (2022).

MANUSCRIPTS UNDER REVIEW OR REVISION

* Denotes equal authorship

2. Basak, A.*, Ortiz-Cordero, C.*, Yiu, S.P.T.*, Allison, S., Sweeting, S., Shan, F., Tong, Y.E., Chorghade, R., Sosa-Guir, A., Alakwe, S.D., Tzouanas, C., Zhu, B., Gabba, A., Yeo, Y.Y., Wui, W., Qiu, H., Chang, Y., Wang, C., Carpenter, E.J., Gupta, S., Lima, G.M., Parmelee, L., Rock, P., Zhang, Y., Derda, R., Burack, W.R., Ma, Q., Jiang, S., and Kiessling, L., “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, *bioRxiv*, <https://doi.org/10.64898/2026.02.23.707224> (2026) (*Under revision at Nat Biotech*)
1. Li, W., Mao, L., Liu, Y., Peng, F., Sachs, N., Wu, W., Yiu, S.P.T., Yan, H., Schroeder, A., Yu, X., Jin, K., Jiang, S., Chen, Z., Loth, M., Gomez, L., Lubo, I., Blank, N., Samarah, L.Z., Basak, A., Cho, Y.W., Chen, C.Y., Kim, D.M., Shalek, A.K., Soto, L.M.S., Rabinowitz, J.D., Reilly, M.P., Qian, X., Thaiss, C.A., Maegdefessel, L., Wang, L., Jiang, S., Deng, Y., and Li, M., “Toward Computationally Complete Spatial Omics”, *bioRxiv*, <https://doi.org/10.64898/2026.03.03.709303> (2026)

MANUSCRIPTS UNDER PREPARATION

* Denotes equal authorship

2. Basak, A.*, Zhu, B.*, Cho, Y.W. *, Han, Y. *, Alakwe, S.D., Parmelee, L., Yiu, S.P.T., Chen, C.Y., Kim, D.M., Wang, X., Kiessling, L., and Shalek, A.K., “Single-cell and spatial landscape of human fungiform taste buds in oral mucosal health and inflammation”
1. Basak, A.*, Cho, Y.W. *, Yiu, S.P.T.*, Shiba, T., Hayath, M., Kiessling, L., Jiang, S., Chen, C.Y., Kim, D.M., and Shalek, A.K., “Epithelial cells regulate immune and microbial populations in oral mucosal inflammatory diseases periodontitis and peri-implantitis”

INVITED TALKS

6. “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Broad Institute Board of Scientific Counsellors Annual Meeting, Cambridge, USA, 2026.
5. “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Single-cell Cancer Biology Gordon Research Conference, Ventura, USA, 2026.
4. “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Broad Institute Chemical Biology Supergroup, Cambridge, USA, 2025.
3. Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Koch Institute Focus Talk, Cambridge, USA, 2025.
2. “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Broad Institute Cell Circuits and Epigenetics Program, Cambridge, USA, 2025.
1. “Decoding oral mucosal gene regulatory circuits for targeted therapy design”, MIT-Merck Symposium, Cambridge, USA, 2024.

POSTER PRESENTATIONS

7. Basak, A.*, Ortiz-Cordero, C.*, Yiu, S.P.T.*, Allison, S., Sweeting, S., Shan, F., Tong, Y.E., Chorghade, R., Sosa-Guir, A., Alakwe, S.D., Tzouanas, C., Zhu, B., Gabba, A., Yeo, Y.Y., Wui, W., Qiu, H., Chang, Y., Wang, C., Carpenter, E.J., Gupta, S., Lima, G.M., Parmelee, L., Rock, P., Zhang, Y., Derda, R., Burack, W.R., Ma, Q., Jiang, S., and Kiessling, L., “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, American Association of Immunology Annual Conference, Boston, USA, 2026.
6. Basak, A.*, Ortiz-Cordero, C.*, Yiu, S.P.T.*, Allison, S., Sweeting, S., Shan, F., Tong, Y.E., Chorghade, R., Sosa-Guir, A., Alakwe, S.D., Tzouanas, C., Zhu, B., Gabba, A., Yeo, Y.Y., Wui, W., Qiu, H., Chang, Y., Wang, C., Carpenter, E.J., Gupta, S., Lima, G.M., Parmelee, L., Rock, P., Zhang, Y., Derda, R., Burack, W.R., Ma, Q., Jiang, S., and Kiessling, L., “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Single-cell Cancer Biology Gordon Research Conference and Seminar, Ventura, USA, 2026.
5. Basak, A.*, Ortiz-Cordero, C.*, Yiu, S.P.T.*, Allison, S., Sweeting, S., Shan, F., Tong, Y.E., Chorghade, R., Sosa-Guir, A., Alakwe, S.D., Tzouanas, C., Zhu, B., Gabba, A., Yeo, Y.Y., Wui, W., Qiu, H., Chang, Y., Wang, C., Carpenter, E.J., Gupta, S., Lima, G.M., Parmelee, L., Rock, P., Zhang, Y., Derda, R., Burack, W.R., Ma, Q., Jiang, S., and Kiessling, L., “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Broad Institute Annual Retreat, Cambridge, USA, 2025.
4. Basak, A.*, Ortiz-Cordero, C.*, Yiu, S.P.T.*, Allison, S., Sweeting, S., Shan, F., Tong, Y.E., Chorghade, R., Sosa-Guir, A., Alakwe, S.D., Tzouanas, C., Zhu, B., Gabba, A., Yeo, Y.Y., Wui, W., Qiu, H., Chang, Y., Wang, C., Carpenter, E.J., Gupta, S., Lima, G.M., Parmelee, L., Rock, P., Zhang, Y., Derda, R., Burack, W.R., Ma, Q., Jiang, S., and Kiessling, L., “Single-Cell and Spatial Methods for Multimodal Functional Glycan Profiling in Tissues”, Carbohydrates Gordon Research Conference and Seminar, New Hampshire, USA, 2025.
3. Basak, A.*, Cho, Y.W. *, Yiu, S.P.T.*, Shiba, T., Hayath, M., Kiessling, L., Jiang, S., Chen, C.Y., Kim, D.M., and Shalek, A.K., “Epithelial cells regulate immune and microbial populations in oral mucosal inflammatory diseases periodontitis and peri-implantitis”, American Association of Immunology Annual Conference, Honolulu, USA, 2025.
2. Basak, A.*, Ortiz-Cordero, C.*, Yiu, S.P.T.*, Allison, S., Tong, Y.E., Chorghade, R., Sosa-Guir, A., Tzouanas, C., Zhu, B., Gabba, A., Yeo, Y.Y., Wui, W., Qiu, H., Chang, Y., Wang, C., Burack, W.R., Ma, Q., Jiang, S., and Kiessling, L., “Developing novel methods to decipher the Glycan Code for Next-Generation Immuno-Therapeutics”, MIT HEALS Collaborative, Cambridge, USA, 2024.

1. Basak, A.*, Ortiz-Cordero, C.*, Yiu, S.P.T.*, Allison, S., Tong, Y.E., Chorghade, R., Sosa-Guir, A., Tzouanas, C., Zhu, B., Gabba, A., Yeo, Y.Y., Wui, W., Qiu, H., Chang, Y., Wang, C., Burack, W.R., Ma, Q., Jiang, S., and Kiessling, L., “Developing novel methods to decipher the Glycan Code for Next-Generation Immuno-Therapeutics”, Ono Pharma Foundation Annual Symposium, Cambridge, USA, 2024.

REFERENCES

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